SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	**Assessment:**	Assessment Tool 1– Pseudocode
Weightage:	30 % of CIA	**Total Marks:**	100
CO3 Statement:	*Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)*		
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Section / Batch:		**Date:**	24-10-2026

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

1. Design a health and self-care mobile app that helps users maintain mental wellbeing with meditations and exercises. This case study talks you through the design process of creating a user- friendly mobile app. Design UI with the following requirements.
2. User Login
3. Register
4. Health Routine Blogs
5. Self-Care Blogs
6.

7.
8. This is an intelligent job assistant that helps users find their ideal job and get to all the way to signing a contract faster and more easily. Design UI with the following requirements.
9. User Login
10. Register Subscription
11. Personal Detail
12. Resume Upload
13. Choosing Job
14.
15.
16. This is a mobile app that can help people get parking slots easily even when they travel beyond their normal routes. Design UI with the following requirements.
17. UserLogin
18. Register Count of Empty Parking
19. Free Parking Slots Display
20. Booking Parking Slots
21.
22.
23. Design a mobile app for Online Body Fitness tips and videos for a good daily routine. Contains the following requirements.
24. User Login
25. Register
26. Fitness Blog
27. Diet Planner
28. Fitness Videos
29. Payment and Subscription

CO3 – AT1: Pseudocode Answers

- ## Health and Self-Care Mobile App

Requirements covered: User login, registration, health routine blogs, self-care blogs, meditation/exercises.

```
BEGIN

    DISPLAY "Welcome to MindCare"

    DISPLAY "1. Register   2. Login"

    INPUT choice
    IF choice = 1 THEN
        INPUT name, email, password

        IF email is valid AND password meets rules THEN SAVE

            user details

            DISPLAY "Registration successful"

        ELSE

            DISPLAY "Enter valid

            details" STOP

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    INPUT email, password

    IF credentials are invalid THEN

        DISPLAY "Login failed"

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DISPLAY "1. Health Routine Blogs 2. Self-Care Blogs 3. Meditation/Exercises" INPUT menuChoice
IF menuChoice = 1 THEN

DISPLAY health routine blog list

INPUT selectedBlog

DISPLAY selected health routine blog

ELSE IF menuChoice = 2 THEN

DISPLAY self-care blog

list INPUT selectedBlog

DISPLAY selected self-care blog

ELSE IF menuChoice = 3 THEN

DISPLAY meditation and exercise options INPUT

selectedActivity

DISPLAY instructions for selected activity ELSE

DISPLAY "Invalid choice"

ENDIF

DISPLAY "Thank you for using MindCare"

END

- ## Intelligent Job Assistant

Requirements covered: Login, registration, subscription, personal details, resume upload, job selection/application.

BEGIN

DISPLAY "Welcome to JobAssist"

DISPLAY "1. Register 2. Login"

INPUT choice
IF choice = 1 THEN

INPUT name, email, password

VALIDATE details

IF details are valid THEN

```
        SAVE account
        DISPLAY "Registration successful"
    ELSE
        DISPLAY "Invalid
        details" STOP
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INPUT email, password

IF credentials are invalid THEN
    DISPLAY "Login failed"
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DISPLAY subscription plans
INPUT plan
IF plan is not free AND payment fails THEN
    DISPLAY "Subscription unsuccessful"
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    SAVE subscription

    INPUT personal details, education, skills, experience, preferred location SAVE personal profile

    DISPLAY "Upload

    resume" INPUT

    resumeFile

    IF resumeFile is valid THEN

        SAVE resumeFile
    ELSE

        DISPLAY "Please upload a valid resume"

        STOP
    ENDIF

    FIND jobs matching skills, experience and preferences DISPLAY

    matching job list
    INPUT selectedJob
    DISPLAY selected job details

    DISPLAY "Apply for this job? (Yes/No)"

    INPUT applyChoice

    IF applyChoice = "Yes"

        THEN SUBMIT job

        application
        DISPLAY "Application submitted"

    ELSE

        DISPLAY "Job not

    applied" ENDIF
END
```

- **Smart Parking Mobile App**

Requirements covered: Login, registration, empty-slot count, free-slot display, parking booking.

```
BEGIN

    DISPLAY "Welcome to SmartPark"

    DISPLAY "1. Register   2. Login"

    INPUT choice
    IF choice = 1 THEN

        INPUT name, email, password, vehicleNumber

        VALIDATE details
        IF details are valid THEN

            SAVE user and vehicle details

            DISPLAY "Registration successful"
        ELSE

            DISPLAY "Invalid

            details" STOP

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    INPUT email, password

    IF credentials are invalid THEN

        DISPLAY "Login failed"

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    INPUT currentLocation or destination

    GET nearby parking locations
    FOR each parkingLocation

        GET totalSlots and occupiedSlots emptySlots =

        totalSlots - occupiedSlots DISPLAY

        parkingLocation, emptySlots
    ENDFOR
    DISPLAY available parking slots


    INPUT selectedParking, slot, bookingTime IF

    selected slot is available THEN
        RESERVE selected slot
        UPDATE slot status to "Booked"


        DISPLAY booking confirmation and booking ID

    ELSE

        DISPLAY "Slot unavailable. Choose another slot" ENDIF
END
```

- ## Online Body Fitness App

Requirements covered: Login, registration, fitness blog, diet planner, fitness videos, payment/subscription.

```
  BEGIN

    DISPLAY "Welcome to FitLife"

    DISPLAY "1. Register   2. Login"

    INPUT choice
    IF choice = 1 THEN

        INPUT name, email, password

        VALIDATE details

        IF details are valid THEN

            SAVE user details

            DISPLAY "Registration successful"

        ELSE

            DISPLAY "Invalid

            details" STOP

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INPUT email, password

IF credentials are invalid THEN

    DISPLAY "Login failed"

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DISPLAY "1. Fitness Blog 2. Diet Planner 3. Fitness Videos 4. Subscription" INPUT menuChoice

IF menuChoice = 1 THEN

    DISPLAY fitness blog

    list INPUT

    selectedBlog

    DISPLAY selected fitness blog

ELSE IF menuChoice = 2 THEN

    INPUT fitnessGoal, foodPreference, dietaryRestrictions CREATE diet

    plan based on user inputs
    DISPLAY diet plan

ELSE IF menuChoice = 3 THEN

    DISPLAY fitness video

    categories INPUT selectedVideo
    PLAY selected video

```
    ELSE IF menuChoice = 4 THEN
        DISPLAY subscription plans and prices
        INPUT selectedPlan, paymentDetails
        PROCESS payment
        IF payment is successful THEN
            ACTIVATE subscription
            DISPLAY "Subscription activated"
        ELSE
            DISPLAY "Payment
        failed" ENDIF
    ELSE
        DISPLAY "Invalid choice"

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